Lesson 11.4 - Matched or Paired Samples

> **Video:** Matched or Paired Samples T-Test - Hypothesis Testing by The Organic Chemistry Tutor
> **Chapter:** 11 - Hypothesis Testing - 2 Samples

When samples are paired

A **matched (paired) sample** measures the **same subjects twice**, for example before and after a treatment. Instead of comparing two separate groups, analyze the **difference** for each subject:

$$x_d = \text{after} - \text{before}$$

Formulas

Quantity	Formula
Mean difference	$\bar{x}_d = \dfrac{\sum x_d}{n}$
SD of the differences	$s_d = \sqrt{\dfrac{\sum (x_d - \bar{x}_d)^2}{n-1}}$ (or `=STDEV.S(...)` in Excel)
Test statistic	$t = \dfrac{\bar{x}_d - \mu_d}{s_d/\sqrt{n}}$, $\text{df} = n - 1$
Confidence interval	$\bar{x}_d \pm t\,\dfrac{s_d}{\sqrt{n}}$

Example - a weight-loss program

Is the program effective at reducing weight? $\alpha = 0.05$, 10 subjects.

Subject	Before (lb)	After (lb)	x_d = after − before
1	185	169	−16
2	192	187	−5
3	206	193	−13
4	177	176	−1
5	225	194	−31
6	168	171	+3
7	256	228	−28

Subject	Before (lb)	After (lb)	x_d = after − before
8	239	217	−22
9	199	204	+5
10	218	195	−23

A negative difference means the subject lost weight.

Hypotheses. An effective program means the mean difference is **negative**:

$$H_0 : \mu_d \geq 0 \qquad H_a : \mu_d < 0 \quad \text{(left-tailed)}$$

Sample statistics:

$$\bar{x}_d = \frac{-131}{10} = -13.1, \qquad s_d = \sqrt{\frac{1526.9}{9}} \approx 13.025$$

Critical value: left-tailed, df $= 9$, $\alpha = 0.05$ gives $t = -1.833$.

Test statistic:

$$t = \frac{-13.1 - 0}{13.025/\sqrt{10}} = \frac{-13.1}{4.119} \approx -3.18$$

$-3.18 < -1.833$, which is in the rejection region. **Reject H_0.** There is sufficient evidence at the 5% level that the program reduces weight.

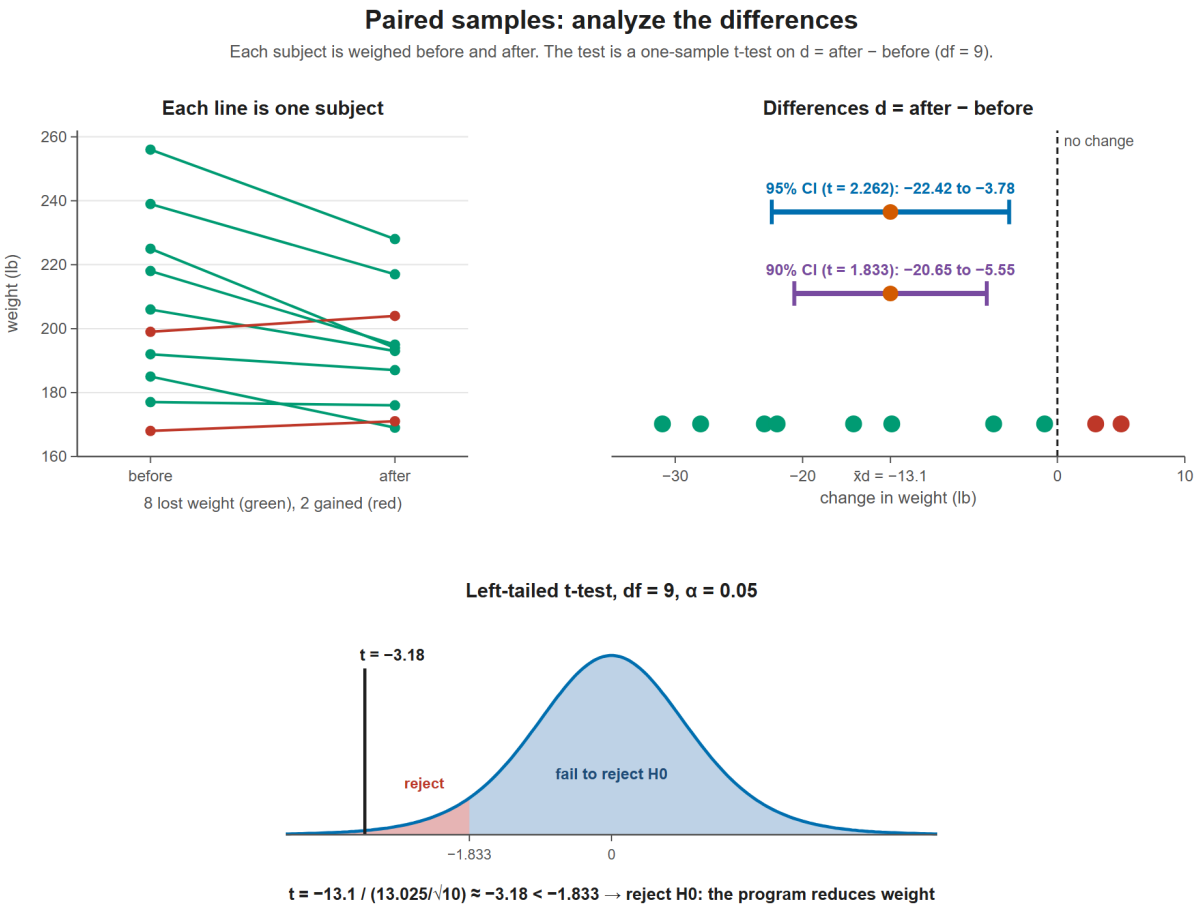
Confidence interval for μ_d

Using $t = 1.833$ (as in the video):

$$-13.1 \pm 1.833\,(4.119) = -13.1 \pm 7.55 \;\Rightarrow\; (-20.65,\ -5.55)$$

> **Correction:** 1.833 is the critical value with **5% in one tail**, so a two-sided interval built from it is a **90%** confidence interval, not 95%. For a **95%** interval, use $t_{0.025,\,9} = 2.262$: $-13.1 \pm 2.262(4.119) = -13.1 \pm 9.32 \Rightarrow (-22.42,\ -3.78)$.

Either way, the whole interval is **below 0**: there are no zero or positive values, which supports rejecting H_0. The margin of error is 7.55 for the 90% interval, or 9.32 for 95%.



Top left: most subjects lost weight. Top right: the ten differences, with the 90% and 95% intervals for the mean difference; both sit entirely below 0. Bottom: t = â□□3.18 is beyond the critical value â□□1.833, so H0 is rejected.

Key takeaways

- Paired data → work with the **differences** x_d and run a one-sample t-test on them.
- df $= n - 1$, where n is the number of **pairs**.
- Choose the direction of H_a to match the claim (e.g. "reduces" → $\mu_d < 0$ when x_d = after − before).
- Match the CI's t to its confidence level: two-sided 95% uses $\alpha/2 = 0.025$.